

How to Apply the Virtual Crystal Approximation (VCA) in VASP

A Practical Tutorial Using $GaAs_{0.5}P_{0.5}$

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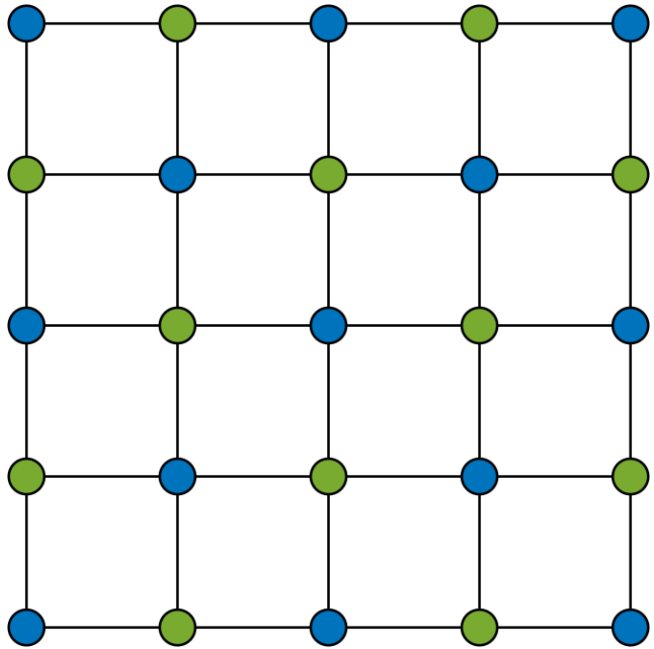
What Will You Learn?

After this tutorial, you will be able to:

- Understand the basic idea of VCA
- Set up a VCA calculation in VASP
- Prepare the required input files
- Analyze the resulting density of states (DOS)

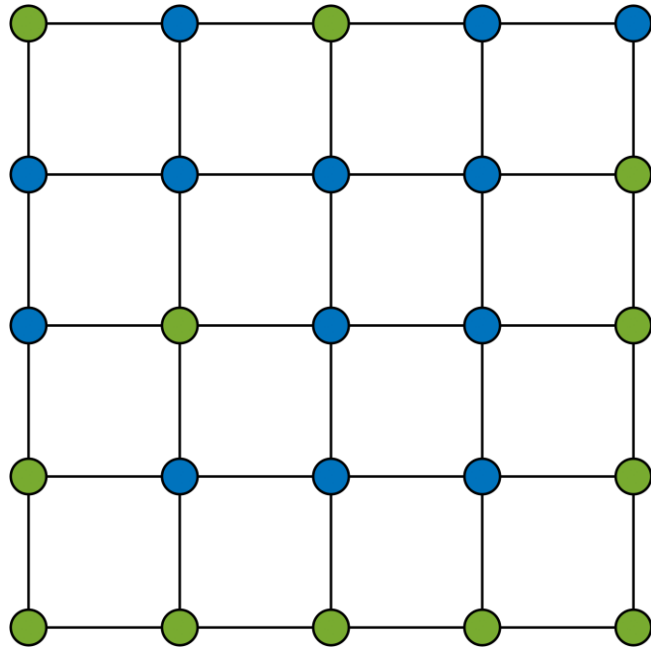
Why Do We Need VCA?

Ordered Crystal



- ✓ Periodic
- ✓ Primitive unit cell sufficient

Random Alloy



- X Non-periodic
- X No primitive cell
- X Large supercells required

Problem

Random alloys lose translational periodicity and require large supercells

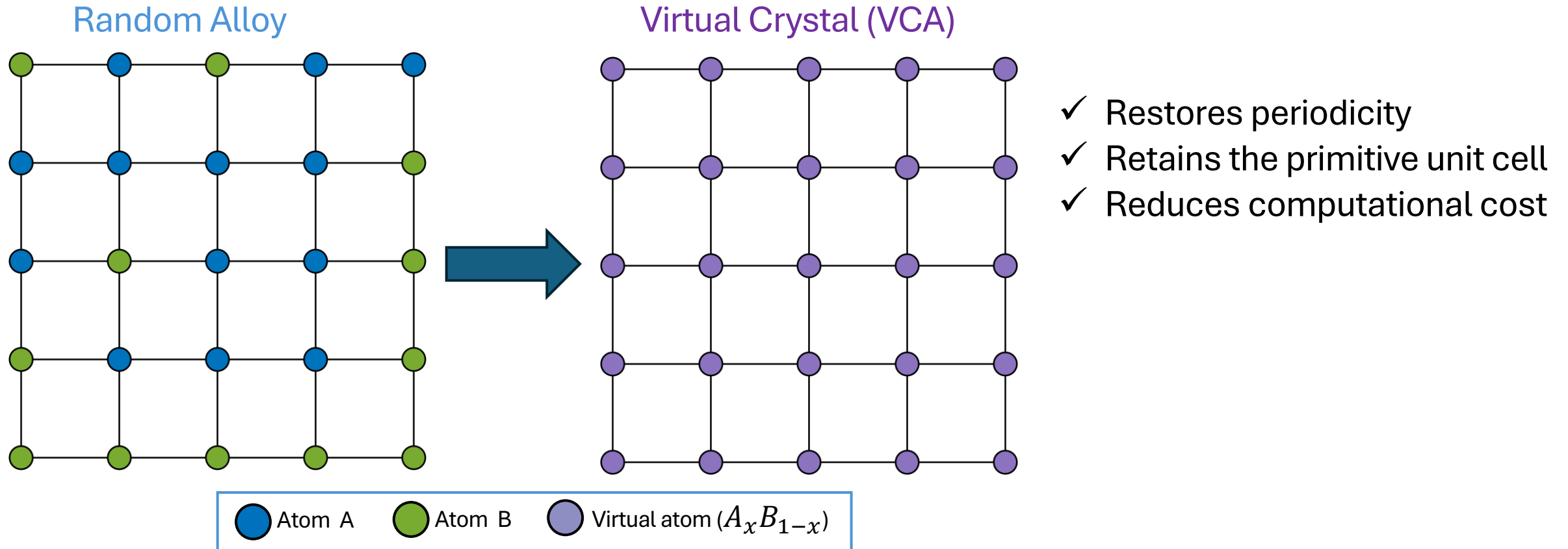
Solution

VCA mathematically restores a configurationally averaged periodic crystal framework, enabling standard primitive cell calculations

What Does VCA Do?

Replace the disordered alloy potential with an ordered, averaged potential

$$V_{VCA} = xV_A + (1 - x)V_B$$

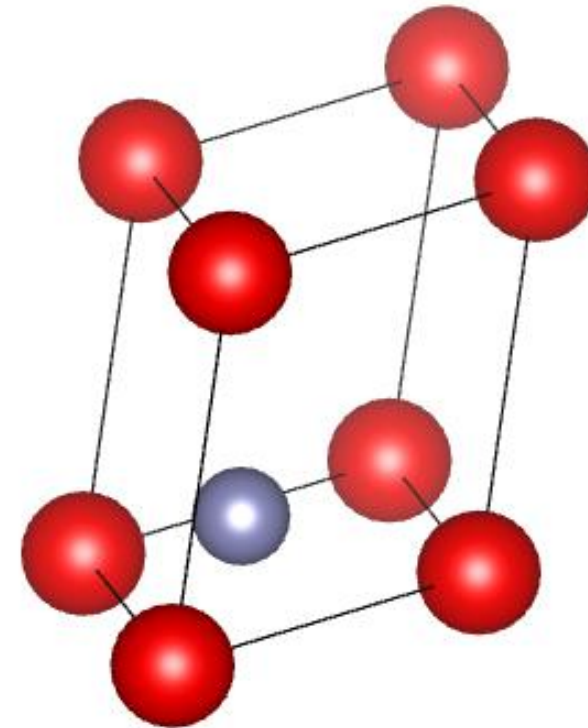
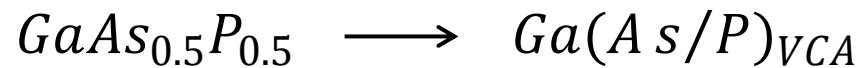


Example System : $GaAs_{0.5}P_{0.5}$

Alloy Composition

- Ga occupies the cation sites
- As and P share the anion sites
- VCA replaces the disordered As/P sites with a uniform virtual atom

VCA Representation



● Ga

● Virtual atom: $(As_{0.5}P_{0.5})_{VCA}$

Primitive zinc-blende unit cell

Computational Workflow:

Structure Relaxation: Optimize atomic positions and lattice parameters



(CONTCAR)



Static Calculation: Generate a converged charge density



(CHGCAR)



DOS Calculation: Compute the density of states

VCA-Implementation in VASP

Step 1: Prepare the POSCAR File

```
GaAs0.5P0.5 ! Bulk System
5.655 ! Lattice parameter (Å) via Vegard's law
0.0000 0.5000 0.5000
0.5000 0.0000 0.5000 ! FCC lattice vectors
0.5000 0.5000 0.0000
Ga As P ! Atomic species (Must match POTCAR)
1 1 1 ! Number of atoms
Direct ! Fractional coordinates mode
0.0000 0.0000 0.0000 ! Ga (Cation site)
0.2500 0.2500 0.2500 ! As (Coincident virtual anion site)
0.2500 0.2500 0.2500 ! P (Coincident virtual anion site)
```

VCA-Implementation in VASP

Step 2: Enable VCA in INCAR

1. Structure Relaxation

System = Bulk $GaAs_{0.5}P_{0.5}$

ISTART = 0

ICHGCAR = 2

ENCUT = 400

ISMEAR = 0

SIGMA = 0.05

ISIF = 3 # Relax ions, cell volume, and cell shape

NSW = 200

IBRION = 2

EDIFF = 1E-5

EDIFFG = -0.01

VCA = 1.0 0.5 0.5 # Mix ratio: 100% Ga, 50% As, 50% P

2. Static Calculation (SCF)

System = Bulk $GaAs_{0.5}P_{0.5}$

ISTART = 0

ICHGCAR = 2

ENCUT = 400

ISMEAR = 0

SIGMA = 0.05

NSW = 0

IBRION = -1

EDIFF = 1E-5

EDIFFG = -0.01

VCA = 1.0 0.5 0.5

VCA-Implementation in VASP

Step 2: Enable VCA in INCAR

3. DOS Calculation

System = Bulk $GaAs_{0.5}P_{0.5}$

ISTART = 1

ICHGCAR = 11 # Read charge density from the previous CHGCAR

ENCUT = 400

ISMEAR = -5 # Tetrahedron method for accurate DOS curves

SIGMA = 0.05

ISIF = 0

NSW = 0

IBRION = -1

EDIFF = 1E-5

EDIFFG = -0.01

VCA = 1.0 0.5 0.5

LORBIT = 11 # Outputs DOSCAR and PROCAR

VCA-Implementation in VASP

Step 3: Prepare KPOINTS and POTCAR

KPOINTS (Monkhorst-Pack Mesh Alignment):

Automatic mesh

0

Monkhorst Pack

6 6 6

0 0 0

POTCAR

Combine the standard PAW-PBE pseudopotentials using your command line terminal:

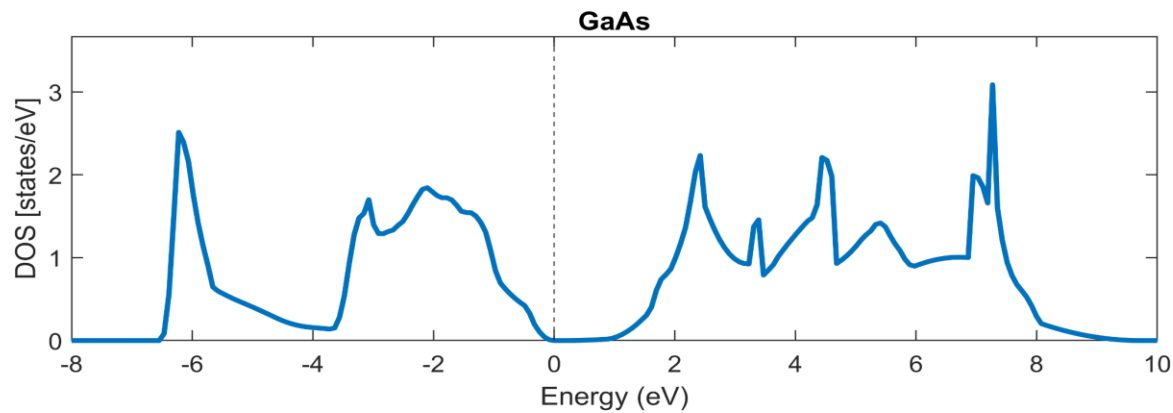
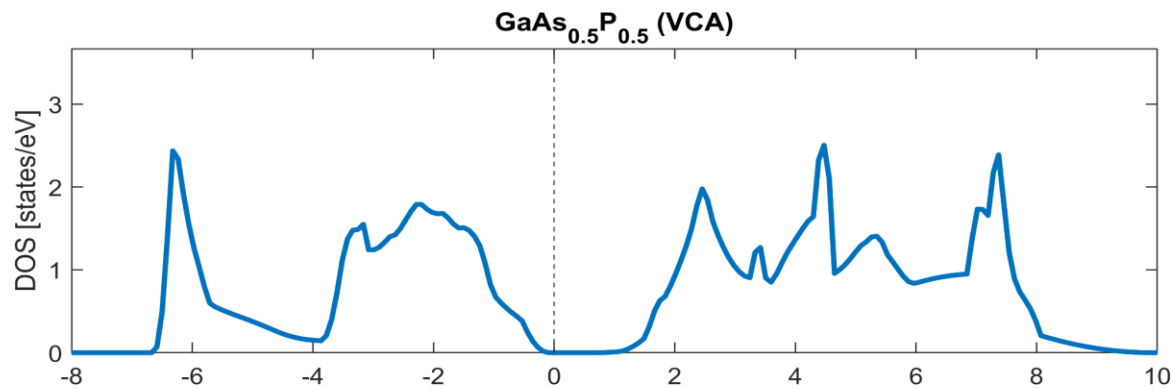
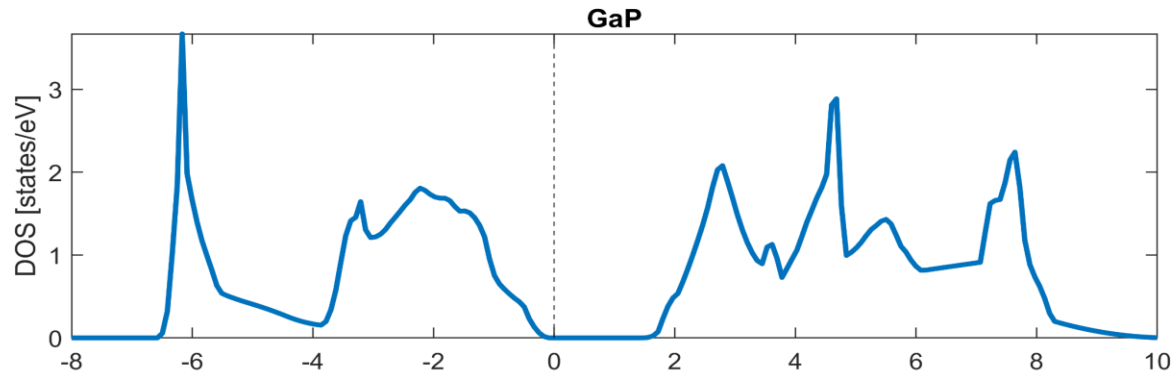
```
cat POTCAR_Ga POTCAR_As
```

```
POTCAR_P > POTCAR
```

Critical VCA Rule:

- The POTCAR file must include all three species (Ga, As, P)
- The file ordering must strictly match the order listed in your POSCAR file!

Results: Density of States (DOS)



Key Observations:

- DOS lies between GaAs and GaP
- Main electronic features are preserved
- Smooth spectrum reflects the averaged crystal potential

SUMMARY

- VCA is an efficient method for modeling substitutional alloys
- VASP supports VCA through the VCA tag
- Only minor modifications to standard VASP inputs are required
- $GaAs_{0.5}P_{0.5}$ can be simulated efficiently using a primitive cell